

Devoir - cyber

INTRODUCTION

J'ai dû réaliser un projet de déploiement réseau.

L'objectif était d'étendre l'architecture réseau existante de l'entreprise en créant un nouveau sous-réseau sécurisé hébergeant un service Nextcloud. Ce service permet aux utilisateurs de stocker et partager des fichiers de manière centralisée au sein du système d'information.

Pour réaliser ce projet, j'ai utilisé une machine virtuelle sous Debian 13 sans interface graphique pour héberger le service, ainsi que le logiciel GNS3 pour simuler l'infrastructure réseau avec un routeur Cisco 3725.

Ce document retrace l'ensemble des étapes que j'ai suivies pour mener à bien ce projet, depuis la conception de la table d'adressage jusqu'à la configuration sécurisée de la machine virtuelle, en respectant toutes les contraintes globales et variables définies dans le cahier des charges.

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Table d'adressage

Équipement	Adresse IP	IP Réseau	1ère Adresse	Dernière adresse	Masque
R1	10.200.120.1				
	10.200.112.1				
	10.200.121.1				
SW1		10.200.120.0	10.200.120.2	10.200.120.254	/24
SW2		10.200.112.0	10.200.112.3	10.200.112.254	/20
SW3		10.200.121.0	10.200.121.3	10.200.121.254	/24
PC Admin	10.200.112.2				
Serveur Nextcloud	10.200.121.2				

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Configuration du routeur R1

```
en
conf t

#SW1
int fa 0/0
description LAN_ZONE_OFFICE
ip address 10.200.120.1 255.255.255.0
no shutdown
exit

ip dhcp pool zone_office
network 10.200.120.0 255.255.255.0
default-router 10.200.120.1
dns-server 8.8.8.8
lease 1
exit

ip dhcp excluded-address 10.200.120.1 10.200.120.2

#SW2
int fa 0/1
description LAN_ZONE_TECH
ip address 10.200.112.1 255.255.240.0
no shutdown
exit

ip dhcp pool zone_technique
network 10.200.112.0 255.255.240.0
default-router 10.200.112.1
dns-server 8.8.8.8
lease 1
exit

ip dhcp excluded-address 10.200.112.1

#SW3
int fa 1/0
description LAN_ZONE_NEXTC
ip address 10.200.121.1 255.255.255.0
no shutdown
exit

ip dhcp
pool zone_nextcloud
network 10.200.121.0 255.255.255.0
default-router 10.200.121.1
dns-server 8.8.8.8
lease 1
exit

ip dhcp excluded-address 10.200.121.1
ip dhcp excluded-address 10.200.121.2

exit
write memory
co ru st
```

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Installation des mises à jour

`sudo apt update`

```
Setting up gnupg-utils (2.4.7-21+deb13u1+b2) ...
Setting up apache2-utils (2.4.66-1~deb13u2) ...
Setting up openssh-sftp-server (1:10.0p1-7+deb13u2) ...
Setting up gpg-agent (2.4.7-21+deb13u1+b2) ...
Setting up openssh-server (1:10.0p1-7+deb13u2) ...
ssh.socket is a disabled or a static unit not running, not starting it.
Setting up mariadb-plugin-provider-lz4 (1:11.8.6-0+deb13u1) ...
Setting up mariadb-plugin-provider-snappy (1:11.8.6-0+deb13u1) ...
Setting up apache2 (2.4.66-1~deb13u2) ...
apache-htcacheclean.service is a disabled or a static unit not running, not starting it.
Setting up gpgsm (2.4.7-21+deb13u1+b2) ...
Setting up inetutils-telnet (2:2.6-3+deb13u3) ...
```

Mise à jour du nouveau nom d'utilisateur

(tiret du bas non-exploité par debian)

```
root@Deb12-Test:~# hostnamectl set-hostname Fileserver_03.lan
root@Deb12-Test:~# hostname
Fileserver03.lan
```

Création des nouveaux utilisateurs

(Connexion en SSH pour plus de facilité)

```
root@Fileserver03:~# adduser nc_admin
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for nc_admin
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
root@Fileserver03:~# useradd -m -s /usr/sbin/nologin cloud_data
```

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Sécurisation des fichiers de configuration ssh

PermitRootLogin no

AllowUsers nc_admin

```
GNU nano 8.4 /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
```

Installation de LAMP

```
root@fileservers03:~# apt install apache2 libapache2-mod-php php-gd php-mysql php-curl php-mbstring ph
p-intl php-gmp php-bcmath php-xml php-zip -y
apache2 is already the newest version (2.4.66-1-deb13u2).
libapache2-mod-php is already the newest version (2:8.4+96).
php-gd is already the newest version (2:8.4+96).
php-mysql is already the newest version (2:8.4+96).
php-curl is already the newest version (2:8.4+96).
php-mbstring is already the newest version (2:8.4+96).
php-xml is already the newest version (2:8.4+96).
php-xml set to manually installed.
php-zip is already the newest version (2:8.4+96).
The following package was automatically installed and is no longer required:
  linux-image-6.12.63+deb13-amd64
Use 'apt autoremove' to remove it.

Installing:
  php-bcmath php-gmp php-intl

Installing dependencies:
  libicu76 php8.4-bcmath php8.4-gmp php8.4-intl

Summary:
  Upgrading: 0, Installing: 7, Removing: 0, Not Upgrading: 0
  Download size: 9938 kB
  Space needed: 39.2 MB / 15.8 GB available

Get:1 http://security.debian.org/debian-security trixie-security/main amd64 php8.4-bcmath amd64 8.4.1
6-1-deb13u1 [26.5 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libicu76 amd64 76.1-4 [9722 kB]
Get:3 http://security.debian.org/debian-security trixie-security/main amd64 php8.4-gmp amd64 8.4.16-1
-deb13u1 [22.2 kB]
Get:4 http://security.debian.org/debian-security trixie-security/main amd64 php8.4-intl amd64 8.4.16-
1-deb13u1 [155 kB]
24% [2 libicu76 1735 kB/9722 kB 18%] 144 kB/s 55s
```

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Installation de MARIADB

```
root@FilesServer03:~# apt install mariadb-server -y
mariadb-server is already the newest version (1:11.8.6-0+deb13u1).
The following package was automatically installed and is no longer required:
  linux-image-6.12.63+deb13-amd64n trixie/main amd64 php-gmp all 2:8.4+96 [3960 B]
Use 'apt autoremove' to remove it. trixie/main amd64 php-intl all 2:8.4+96 [3964 B]
Fetched 9938 kB in 1min 1s (164 kB/s)
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

Création de la base de données Nextcloud

```
MariaDB [(none)]> CREATE DATABASE nextcloud;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> CREATE USER 'cloud_data'@'localhost' IDENTIFIED BY 'debian';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON nextcloud.* TO 'cloud_data'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> EXIT;
Bye
```

Installation de Nextcloud depuis la documentation officielle

```
root@FilesServer03:/var/www/html# wget -4 https://download.nextcloud.com/server/releases/latest.tar.bz2
--2026-04-13 19:47:19-- https://download.nextcloud.com/server/releases/latest.tar.bz2
Resolving download.nextcloud.com (download.nextcloud.com)... 5.9.202.145
Connecting to download.nextcloud.com (download.nextcloud.com)[5.9.202.145]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 274858046 (262M) [application/x-bzip2]
Saving to: 'latest.tar.bz2'

latest.tar.bz2      0%[                  ] 343.59K  20.0KB/s   eta 3h 45m
```

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Téléchargement depuis le PC hôte pour ensuite faire un partage de fichiers via WinScp

/home/debian/				
Nom	Taille	Date de modification	Droits	Proprié...
..		13/04/2026 19:26:39	rw-r--r--	root
Desktop		16/03/2024 17:20:28	rw-r--r--	debian
Documents		16/03/2024 17:20:28	rw-r--r--	debian
Downloads		16/03/2024 17:20:28	rw-r--r--	debian
Music		16/03/2024 17:20:28	rw-r--r--	debian
Pictures		16/03/2024 17:20:28	rw-r--r--	debian
Public		16/03/2024 17:20:28	rw-r--r--	debian
Templates		16/03/2024 17:20:28	rw-r--r--	debian
Videos		16/03/2024 17:20:28	rw-r--r--	debian
latest.tar.bz2	268 417 KB	13/04/2026 19:50:30	rw-rw-r--	debian

cloud_data		13/04/2026 19:26:39	rwX-----	cloud_...
debian		18/01/2026 21:44:58	rwX-----	debian
nc_admin		13/04/2026 19:26:09	rwX-----	nc_ad...

Déplacer vers la page html

```
root@Fileserver03:~# cd /home/debian
root@Fileserver03:/home/debian# ls
Desktop Documents Downloads Music Pictures Public Templates Videos latest.tar.bz2
root@Fileserver03:/home/debian# mv latest.tar.bz2 /var/www/html
root@Fileserver03:/home/debian#
```

Extraction du fichier Nextcloud, puis attribution des permissions

```
root@Fileserver03:/var/www/html# tar -xjf latest.tar.bz2
root@Fileserver03:/var/www/html# chown -R www-data:www-data /var/www/html/nextcloud
root@Fileserver03:/var/www/html# chown -R 755 /var/www/html/nextcloud
root@Fileserver03:~# chown -R 775 /var/www/html/nextcloud/
root@Fileserver03:~# chmod -R 775 /var/www/html/nextcloud/
root@Fileserver03:~#
```


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Installation de fail2ban

```
sudo apt install fail2ban -y
```

```
Installing:
  fail2ban

Installing dependencies:
  python3-pyasyncore python3-pyinotify python3-systemd whois

Suggested packages:
  system-log-daemon monit sqlite3 python-pyinotify-doc

Summary:
  Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 0
  Download size: 615 kB
  Space needed: 3014 kB / 15.4 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 python3-systemd amd64 235-1+b6 [40.8 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 fail2ban all 1.1.0-8 [466 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 python3-pyasyncore all 1.0.2-3 [10.1 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 python3-pyinotify all 0.9.6-5 [24.9 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 whois amd64 5.6.3 [72.4 kB]
Fetched 615 kB in 4s (169 kB/s)
Selecting previously unselected package python3-systemd.
(Reading database ... 45745 files and directories currently installed.)
Preparing to unpack .../python3-systemd_235-1+b6_amd64.deb ...
Unpacking python3-systemd (235-1+b6) ...
Selecting previously unselected package fail2ban.
Preparing to unpack .../fail2ban_1.1.0-8_all.deb ...
Unpacking fail2ban (1.1.0-8) ...
Selecting previously unselected package python3-pyasyncore.
Preparing to unpack .../python3-pyasyncore_1.0.2-3_all.deb ...
Unpacking python3-pyasyncore (1.0.2-3) ...
Selecting previously unselected package python3-pyinotify.
Preparing to unpack .../python3-pyinotify_0.9.6-5_all.deb ...
Unpacking python3-pyinotify (0.9.6-5) ...
Selecting previously unselected package whois.
Preparing to unpack .../archives/whois_5.6.3_amd64.deb ...
Unpacking whois (5.6.3) ...
Setting up whois (5.6.3) ...
Setting up python3-pyasyncore (1.0.2-3) ...
Setting up python3-systemd (235-1+b6)
```

Configuration du banissement

```
GNU nano 8.4 /etc/fail2ban/jail.local *
[sshd]
enabled = true
maxretry = 2
findtime = 300
bantime = 172800
```


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Installation du pare-feu UFW

```
root@FilesServer03:~# apt install ufw -y
The following package was automatically installed and is no longer required:
  linux-image-6.12.63+deb13-amd64
Use 'apt autoremove' to remove it.

Installing:
  ufw

Installing dependencies:
  iptables libip6tc2 libnetfilter-conntrack3 libnftnl0

Suggested packages:
  firewalld rsyslog

Summary:
  Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 0
  Download size: 607 kB
  Space needed: 3611 kB / 15.4 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 libip6tc2 amd64 1.8.11-2 [20.3 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libnftnl0 amd64 1.0.2-3 [14.4 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libnetfilter-conntrack3 amd64 1.1.0-1 [42.1 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 iptables amd64 1.8.11-2 [361 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 ufw all 0.36.2-9 [169 kB]
Fetched 607 kB in 3s (193 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libip6tc2:amd64.
(Reading database ... 46280 files and directories currently installed.)
Preparing to unpack .../libip6tc2_1.8.11-2_amd64.deb...
```

Ouverture des ports nécessaires Bloquage de l'adresse ip du SW2

```
root@FilesServer03:~# ufw allow 80/tcp
Skipping adding existing rule
root@FilesServer03:~# ufw allow 443/tcp
Rules updated
Rules updated (v6)
root@FilesServer03:~# ufw allow from 10.200.112.2 to any port 22
Rules updated
root@FilesServer03:~# ufw deny from 10.200.112.0/20
Rules updated
root@FilesServer03:~# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@FilesServer03:~# exit
```

Activation des règles sudo ufw enable

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CONFIGURATION DE L'IP STATIQUE

Configuration de l'adresse IP

```
GNU nano 8.4 /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto enp0s3
iface enp0s3 inet static
    address 10.200.121.2
    netmask 255.255.255.0
    gateway 10.200.121.1
```

Vérification de l'adresse IP et du nom de l'hôte

```
GNU nano 8.4 /etc/hosts *
127.0.0.1    localhost
127.0.1.1    Fileserver_03.lan    Fileserver_03

# The following lines are desirable for IPv6 capable hosts
::1          localhost ip6-localhost ip6-loopback
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters

10.200.121.2 Fileserver_03.lan
```

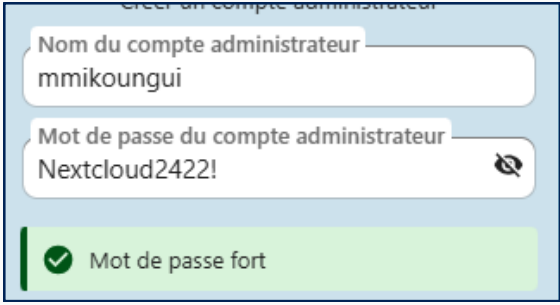
```
Machine  Ecran  Entrée  Fenêtres  Aide
GNU nano 8.4 nextcloud.conf
<VirtualHost *:80>
    DocumentRoot /var/www/html/nextcloud
    ServerName Fileserver_03.lan

    <Directory /var/www/html/nextcloud>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

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CONNEXION À NEXTCLOUD

Création du nouveau compte administrateur



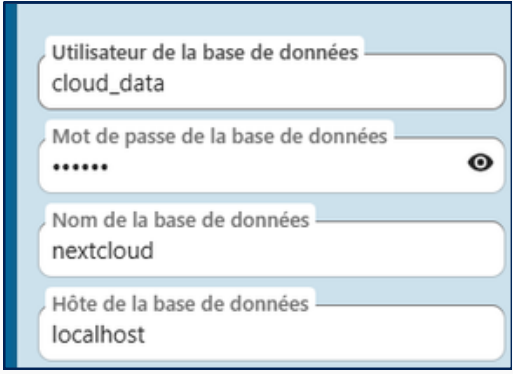
Créer un compte administrateur

Nom du compte administrateur
mmikoungui

Mot de passe du compte administrateur
Nextcloud2422!

✓ Mot de passe fort

Création du compte utilisateur `cloud_data`



Utilisateur de la base de données
cloud_data

Mot de passe de la base de données

Nom de la base de données
nextcloud

Hôte de la base de données
localhost

Accès à l'interface Nextcloud

